

**Indian Institute of Technology Gandhinagar**  
**ES116 Principles and Applications of Electrical Engineering**  
**Sem II, 2025-2026, Date: 28/03/2026**  
**Quiz-II, Duration: 90 mins, Maximum Points: 40**

**Instructions:**

1. This assessment accounts for max. of 10 marks towards your total marks.
2. It is recommended that you solve the questions in the given order.
3. Show all your work clearly. Partial credit for correct reasoning.

You are part of a team designing a Brain-Computer Interface (BCI) system. Your goal is to design analog circuits that can read weak brain signals, process them, and convert them into a form usable by a digital microcontroller. Each stage of the system corresponds to one part of the question.

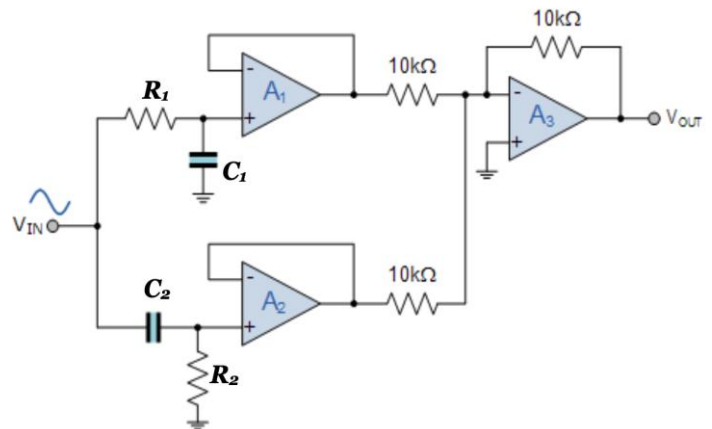
Q1. A pair of electrodes measures a differential brain signal, but it is heavily contaminated by 50 Hz power-line interference. The voltages at the two electrodes are:  $v_1(t) = 0.002 \sin(2\pi \cdot 4t) \text{ V} + 0.001 \sin(2\pi \cdot 50t) \text{ V}$ , and  $v_2(t) = -0.002 \sin(2\pi \cdot 4t) \text{ V} + 0.001 \sin(2\pi \cdot 50t) \text{ V}$ . This signal is fed into a 3-Op-Amp Instrumentation Amplifier. The measured output of the amplifier is:  $v_{\text{out}}(t) = 4 \sin(2\pi \cdot 4t) \text{ V} + 0.001 \sin(2\pi \cdot 50t) \text{ V}$ .

- (a) (2 points) Calculate the differential gain ( $A_d$ ) and the common-mode gain ( $A_{\text{cm}}$ ) of the amplifier.
- (b) (1 point) Calculate the corresponding CMRR in dB.
- (c) (4 points) Based on the behavior observed above, draw the corresponding schematic of the Instrumentation Amplifier and guess the values of the resistors in the design. It is given that all four resistors in the difference amplifier have the exact same value.

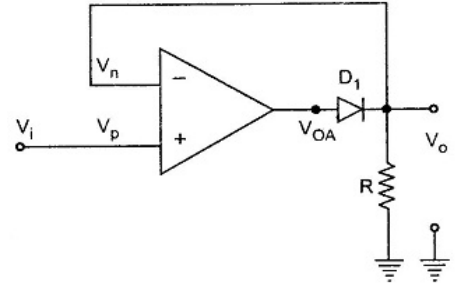
Q2. To eliminate the 50 Hz noise, the  $v_{\text{out}}(t)$  signal from Q1 is fed into the active filter circuit shown here.

- a) (4 points) Derive the overall transfer function of the circuit,  $V_{\text{OUT}}(j\omega)/V_{\text{IN}}(j\omega)$ , in terms of  $R_1$ ,  $C_1$ ,  $R_2$ ,  $C_2$ , and  $\omega$ .
- b) (2 points) The system requires the  $A_1$  branch to have a cutoff frequency of  $f_1 = 15.9 \text{ Hz}$  and the  $A_2$  branch to have a cutoff frequency of  $f_2 = 159 \text{ Hz}$ . Given that  $C_1 = C_2 = 1 \mu\text{F}$ , calculate the required values for resistors  $R_1$  and  $R_2$ . [Hint:  $2\pi \cdot 15.9 \sim 100$ ]
- c) (3 points) Calculate the gain magnitude at  $f \rightarrow 0 \text{ Hz}$ ,  $f \rightarrow \infty$ , and  $f = \sqrt{f_1 f_2} \sim 50 \text{ Hz}$ .

- d) (2 points) Briefly explain why the engineers chose this specific combination of cutoff frequencies for the Brain-Computer Interface (BCI) system.

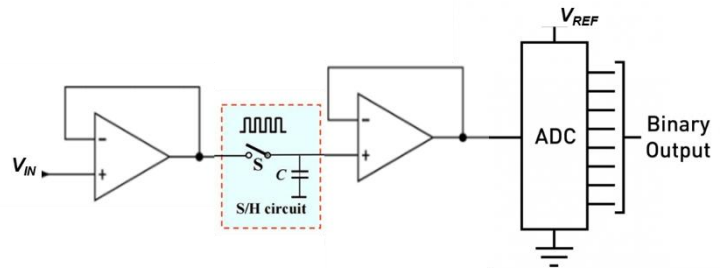


Q3. To ensure compatibility with the next digital stage, the signal is passed through the circuit shown here. A constant 0.7 V forward drop can be assumed across  $D_1$  when it is conducting and Open loop Op-Amp Gain  $\rightarrow \infty$ .



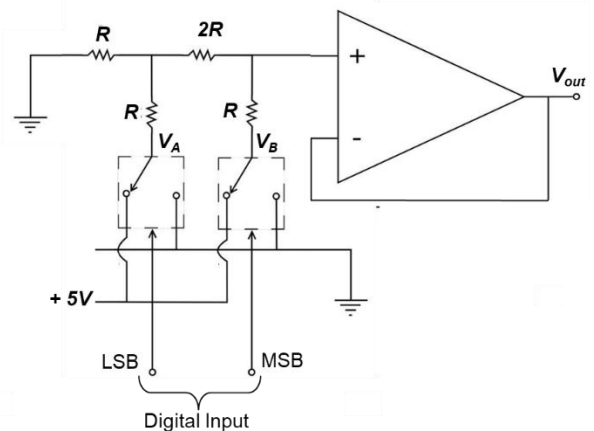
- (4 point) Draw the output waveform  $V_o(t)$  for one complete cycle of the input signal ( $V_i = 4 \sin(2\pi \cdot 4t)$  V). Clearly label the peak voltage and explain how the circuit achieves a specific output value.
- (2 points) Why would you prefer this circuit over a simple diode rectifier (just a diode and a resistor) for this application? Briefly justify.

Q4. A brainwave signal from Q3 is passed through a Sample-and-Hold (S/H) circuit into a 10-bit ADC ( $V_{ref} = 5$  V) as shown in the circuit. The system must be able to detect voltage changes as small as 2 mV.



- (1 point) Explain ADC resolution in the context of this system.
- (2 point) At a sampling instant ( $t=t_s$ ), the switch in the sample-and-hold circuit briefly closes. What happens to the output of the S/H circuit immediately after the switch opens? Why is a sample-and-hold circuit necessary before an ADC?
- (1 point) Calculate the exact step size ( $V_{LSB}$ ) of the 10-bit ADC.
- (2 point) Determine the minimum number of bits required to meet the 2 mV specification. Show your math.

Q5. The microcontroller must output an analog voltage to control the grip strength of a robotic hand. A junior engineer attempts to build a 2-bit R-2R ladder DAC using logic levels of 0 V (Logic 0) and 5 V (Logic 1), and implements the circuit as shown here.



- (6 points) Assuming an ideal Op-Amp, derive the exact equation for  $V_{out}$  in terms of the input voltages  $V_A$  and  $V_B$ .
- (2 points) Calculate the outputs when the system sends digital inputs codes of **10** and **01**. Is the circuit binary weighted? Justify.
- (2 point) Explain why the two input signals ( $V_A$  and  $V_B$ ) interfere with each other when connected to the non-inverting (+) pin. What key property of a standard inverting summing is missing in this circuit?